

# Terafab Advanced Semiconductor Packaging Facility

Technical Setup Plan for the Packaging Module at Giga Texas (Austin) and Gibbons Creek (Grimes County, Texas)

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CLASSIFICATION: INTERNAL TECHNICAL USE

# Terafab Advanced Semiconductor Packaging Facility

## Technical Setup Plan

This document specifies how to stand up the **chip packaging module** of Terafab. It is limited to the public record for Terafab, the packaging processes Intel Foundry already runs as the announced partner, and Texas / U.S. requirements that apply to a semiconductor packaging plant at Gibbons Creek. Where Terafab has not published a number or a tool list, that gap is stated. It is not filled in.

|                      |                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DOCUMENT TITLE       | Terafab Advanced Semiconductor Packaging Facility — Technical Setup Plan                                                                                                                                                                      |
| DOCUMENT NUMBER      | TF-PKG-TSP-001                                                                                                                                                                                                                                |
| REVISION             | A                                                                                                                                                                                                                                             |
| DATE                 | 12 August 2026                                                                                                                                                                                                                                |
| SUBJECT SITES        | Giga Texas North Campus, Austin, Texas (precursor); Gibbons Creek Reservoir, Grimes County, Texas (volume plant)                                                                                                                              |
| SCOPE                | Advanced packaging facility setup only (EMIB / Foveros / EMIB-T class). Front-end lithography and memory-fab process of record are out of scope except where they interface to package.                                                       |
| BASIS OF RECORD      | SpaceX Terafab update; Grimes County tax-abatement and reinvestment-zone documents; Texas Governor / TEF announcement (6 August 2026); Intel Foundry packaging publications (Fab 9, EMIB, Foveros, EMIB-T); TCEQ 30 TAC §106.225; CWA / RCRA. |
| NOT A SUBSTITUTE FOR | Intel process-of-record transfer packages, TCEQ/USACE applications, or unreleased Terafab tool BOMs.                                                                                                                                          |

**Method.** Confirmed Terafab decisions are stated as facts. Intel’s production packaging sequence is used because Intel is the announced packaging partner and there is no merchant Foveros OSAT. Facility and permit requirements are those that apply in Texas to this class of plant. Unreleased Terafab quantities (cleanroom area, tool counts, HBM vendor) are listed as unknown.

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## 0. What is actually locked

| ITEM                              | RECORD                                                                                                                                                                                                                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Owners                            | Tesla and SpaceX. Intel joined April 2026 as manufacturing/package partner. xAI is in the original announcement.                                                                                                                                                        |
| Products                          | Tesla AI5 (pilot), Optimus and Cybercab edge-inference silicon, SpaceX orbital data-center processors, Starlink-related silicon. Logic + memory + package + test in one plant.                                                                                          |
| Precursor                         | Tesla Advanced Technology Fabrication on Giga Texas North Campus, Austin. Ground broken April 2026. Purpose: iterate mask → fab → package → test on one site. Musk later said pilot volume is a few thousand wafers/month, not 100k. AI5 small-batch 2026, volume 2027. |
| Full plant                        | Gibbons Creek Reservoir, Grimes County, Texas (former Gibbons Creek Steam Electric Station), about 15–20 miles from Texas A&M. Reinvestment zone greater than 22,000 acres; reservoir about 4,000 acres.                                                                |
| Phase 1                           | \$16.8 billion (SpaceX/Tesla). 3,000 permanent jobs. Construction 2026. One public record puts Anderson-Shiro phase-1 capex at \$10.3 billion and construction complete 2028. Full campus target about 100 million ft <sup>2</sup> .                                    |
| Incentives                        | Grimes County tax abatement: \$10 million upfront + \$20 million/year for 35 years. Anderson-Shiro CISD and Iola ISD JETI deals signed (50% M&O, 100% I&S). \$30 million Texas Enterprise Fund.                                                                         |
| Site utilities (county agreement) | Not on ERCOT. On-site natural-gas generation. Industrial water from Gibbons Creek Reservoir, not groundwater (30-day notice if they later apply to drill). On-site WWTP, own fire/medical.                                                                              |
| Packaging tech (partner)          | Intel EMIB, Foveros, and EMIB-T (TSV in the bridge for vertical power to ASIC + HBM). Intel volume packaging today: Chandler AZ (ATTD), Rio Rancho NM Fab 9, Oregon, Malaysia.                                                                                          |
| Project lead (SpaceX)             | John Federspiel (public meetings). Terafab spokesperson at the August 2026 resident meeting: Riley Trennell.                                                                                                                                                            |

Table 1. Public-record facts used as the baseline for this plan.

Node statements conflict in public sources (Musk: **14A** at full scale; other coverage: **18A**). Packaging setup cannot wait for that to settle. The package line must accept **multi-node chiplets** (Intel's stated EMIB advantage).

## 1. Dual-site packaging, not one building on day one

**Austin (now).** The Giga Texas research fab is the only packaging line that can exist in 2026–2027. It exists to close the loop Musk described: make a chip, package it, test it, revise the mask,

without shipping wafers to an OSAT. That is the same reason Intel put Foveros next to fab tools in New Mexico: FOUP-to-FOUP, not air freight to Penang.

**Grimes County (phase 1, through 2028).** This is the volume packaging hall, co-located with logic and memory so HBM and compute tiles never leave the campus. SpaceX's own update: manufacturing, packaging, and testing of advanced logic and memory in one location to cut cycle time.

Do not size the Grimes packaging cleanroom as 100 million ft<sup>2</sup>. That figure is the **full campus**. Intel's comparable U.S. advanced-packaging floors (Fab 9, Chandler ATTD) are fab-like bays measured in tens of thousands of square meters of ISO-controlled space, not the whole building.

## 2. Package architecture the products force

Three product classes, three reliability stacks, one assembly flow:

1. **Automotive / robot (Tesla AI5, Optimus, Cybercab)** — AEC-Q100 (and related AEC-Q) plus Tesla's own PPAP. High-volume, cost-sensitive, thermal cycling and humidity.
2. **Orbital compute (SpaceX)** — radiation, thermal vacuum, outgassing (NASA / ECSS materials lists), and parts screening in the MIL-STD-883 / NASA EEE-INST-002 family. This is why packaging is not a merchant OSAT job.
3. **HBM + logic MCM** — EMIB-T exists specifically to feed power through the silicon bridge via TSVs to ASIC and HBM. That is Intel's 2026 packaging story for AI, not a conventional FCBGA.

Intel's public process family that matches this mix:

- **EMIB 2.5D** — silicon bridges embedded in the organic substrate; dies sit side-by-side. Avoids a full silicon interposer (TSMC CoWoS-S style).
- **EMIB-T** — same bridges with TSVs for vertical power and high-speed D2D. IRPS 2026 paper: first-generation EMIB-T reliability characterization. Public analyst commentary: about 90% tech-validation yield; 98% is the hard ramp. Intel CFO: EMIB-T fab rollout in 2026.
- **Foveros** — chip-on-wafer on a base die/interposer: chip attach → thermal “bake” → capillary/epoxy underfill → water-cooled dicing. Documented on Intel Fab 9 by principal engineer Chris Seibert.
- **Foveros Direct 3D** — Cu–Cu hybrid bond, sub-10 μm pitch, no microbump, no underfill at the bond interface (Intel November 2025 brief). Tooling is the industry bottleneck (BESI–Applied, TEL).
- **EMIB 3.5D** — Foveros stack + EMIB bridges in one package.

Terafab has not published which of these is first silicon. The **minimum** line that matches the announced products is **EMIB or EMIB-T + HBM attach + organic substrate + lid/TIM + test**. Foveros Direct is the scale-up path, gated by hybrid-bond tools.

### 3. Process flow to install (Intel's actual sequence)

Copy the Fab 9 / ATTD flow. Do not invent a new one.

#### Incoming

- Known-good-die (KGD) from the on-site logic fab and memory line (or qualified HBM from the memory partners Intel already uses). Incoming inspection: bump/pad metrology, warpage, ID.
- Organic substrates: Intel's long-standing substrate supplier is **Ibiden**; also **Unimicron** and **AT&S** in the public Foveros supply map. EMIB bridges are **embedded in the substrate** before die attach — substrate quality is the first yield gate.

#### Wafer-level / chip-on-wafer (Foveros path)

1. Chip-attach tool (microbump or hybrid).
2. Reflow / bond anneal ("oven").
3. Underfill / epoxy (not used at the Foveros Direct bond interface).
4. AMHS in FOUPs on OHT (Intel's published Fab 9 description).
5. Water-cooled blade dicing to singulate packages.

#### Substrate / EMIB path

1. Receive EMIB-embedded substrates.
2. Flux, die attach of compute tiles and HBM stacks (fine-pitch flip-chip bonders: industry tools for this class are ASMPT, BESI, Kulicke & Soffa).
3. Reflow, underfill (Henkel and other capillary underfills used in this industry).
4. Lid / heat spreader, TIM (Indium Corp, Honeywell, Dow appear in the same public materials map).
5. Ball attach, singulation, 3D X-ray / CSAM.

#### Test

- Wafer sort before package (on-site fab).
- Package test: ATE (Advantest / Teradyne class) for HBM training, HBM PHY, and high-current power.
- Burn-in, HTOL, HAST, TC for AEC-Q100.
- Space lot: radiation (TID, SEE) at a qualified lab; thermal vacuum; residual gas analysis for outgassing. Terafab has not named the rad-test house.

### 4. Facility systems the packaging bay needs

These are not optional. They are how Intel and every U.S. advanced-packaging plant of this class is built (NIST CHIPS PEA; ISO 14644; SEMI).

## Cleanroom

- Advanced packaging is not a Class 100k OSAT hall. Foveros/EMIB-T is run in fab-like space: ISO 14644-1 **ISO 5–6** at the bonders, ISO 7 in service aisles, ULPA/HEPA recirculation, FFUs, raised floor or waffle slab, mini-environments on tools.
- Construction labor in cleanroom-adjacent phases needs ISO gowning certification. Trennell already told Grimes County millwrights are the high-demand trade.

## AMHS

FOUP + OHT if Foveros stays wafer-level. Substrate lines use magazine/JEDEC trays and AGVs. Mixing both is normal at Intel ATTD.

## UPW / wastewater

Dicing, wet clean, and CMP-like steps (hybrid-bond surface prep) consume ultrapure water. SpaceX committed: reservoir water in, on-site WWTP, reuse. That is the same pattern Amkor's CHIPS Peoria deal uses (reclaim + pretreatment + return). Fluoride, copper, and organics from dicing/underfill are the packaging-specific waste streams. RCRA hazardous waste for solvents, flux, and resist.

## Chemicals / gases

Flux, underfill, molding compound, solvents, forming gas for reflow, CDA, vacuum. Hybrid bonding adds plasma activation gases. TCEQ 30 TAC 106.225 (semiconductor manufacturing PBR) plus 106.4 general conditions: 24-month material-usage records, ESL/GLC limits. A plant this size will almost certainly need an NSR / Title V path rather than PBR-only; 106.225 is written for modifications on a site that already has a commission permit.

## Power and cooling

County documents: self-generated natural gas, not ERCOT. Packaging tools are modest versus EUV, but ATE, reflow, and chillers are continuous loads. Process cooling water and CDA are the packaging utilities that fail first if the gas plant is late — sequence the utility island ahead of tool move-in.

## ESD / vibration

IEC 61340-5-1 for handling. Hybrid bonders need vibration isolation comparable to litho (tool OEM spec, not a guess).

# 5. Permits and compliance (Texas and federal)

Do these in parallel with design. They are the critical path more often than the bonder PO.

- **Air:** TCEQ NSR; 30 TAC 106.225 if any PBR is used; Title V if major. Keep 24-month chemical logs.

- **Water:** TCEQ TPDES / industrial wastewater for the on-site plant; CWA 402; CWA 404 if reservoir intake/outfall or wetlands in the 22,000-acre zone (USACE Galveston District).
- **Waste:** RCRA generator ID; TCEQ industrial hazardous waste.
- **County agreement:** no groundwater without 30-day notice; own fire station and WWTP; not on ERCOT.
- **Export:** ITAR/EAR for space processors and radiation-hardened flows. Packaging travelers and yield data are export-controlled if the die is.
- **Automotive:** IATF 16949 on the Tesla package line if AI5/Optimus go through automotive PPAP.
- **CHIPS Act:** Terafab's published state deal is TEF + JETI, not a named CHIPS award. If Commerce money is later used, NIST PEA conditions (NEPA, CWA, CAA, RCRA) apply. Do not assume CHIPS funding exists until it is awarded.

## 6. Supply chain that is real, not theoretical

| INPUT                  | WHO ACTUALLY SUPPLIES THIS CLASS OF LINE                                                                                                                                                          |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMIB/Foveros process   | Intel Foundry (captive). There is no merchant Foveros OSAT. Terafab either runs Intel process under license/partnership on site, or ships to Chandler/Fab 9 until the Texas line is qualified.    |
| Hybrid bonders         | BESI–Applied Materials, Tokyo Electron. Industry capacity is tight. Tesla is already recruiting Taiwan packaging engineers (CoWoS/SoIC experience) — that is the labor market for this equipment. |
| Microbump bonders      | ASMPT, BESI, K&S.                                                                                                                                                                                 |
| Substrates             | Ibiden (Intel's historical primary), Unimicron, AT&S. ABF substrate lead time is a known industry constraint.                                                                                     |
| HBM                    | Not disclosed for Terafab. EMIB-T's purpose is HBM power delivery; HBM allocation is a separate commercial contract.                                                                              |
| TIM / lids / underfill | Honeywell, Dow, Indium, Henkel (public Foveros materials map).                                                                                                                                    |
| ATE                    | Advantest, Teradyne (industry standard for HBM/AI packages).                                                                                                                                      |

Table 2. Supply base for an EMIB/Foveros-class packaging line.

Do not plan on TSMC CoWoS capacity. Public reporting is that CoWoS is allocated through at least 2028. That is a stated reason Intel packaging is in the Terafab deal.

## 7. Workforce

- Phase 1: 3,000 permanent jobs, full range (fab engineers through food service). SpaceX: 60–80% local at other Texas sites.
- Packaging-specific hires that cannot be local on day one: hybrid-bond process, EMIB substrate, HBM attach, yield. Tesla is recruiting those in Taiwan.
- Construction: millwrights, ISO-certified cleanroom installers, mechanical/electrical for UPW and CDA.
- Pipeline: Texas A&M Semiconductor Institute is 15–20 miles away (Steve Putna quoted on Terafab). Use it for technicians; do not treat it as a substitute for Intel process-transfer staff.

## 8. Sequence that matches the public dates

| WHEN            | PACKAGING WORK                                                                                                                                                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Now<br>(Austin) | Finish Giga Texas North Campus package/test cell. Qualify AI5 package on Intel's current EMIB/Foveros flow (likely with Intel U.S. sites as the golden line). Small-batch 2026.                                                                                                     |
| 2026            | Grimes: site civil, gas plant, reservoir intake, WWTP, fire station. Place long-lead POs: hybrid bonders, high-end flip-chip, ATE, UPW. Start TCEQ/USACE applications. Transfer Intel process of record (POR) and spec the cleanroom to Fab 9/Chandler, not to a conventional OSAT. |
| 2027            | Austin volume for AI5. Grimes cleanroom dry-in, tool install, copy-exact matching to Intel. First packaged lots are correlation lots against Chandler/Fab 9, not customer ships.                                                                                                    |
| 2028            | Phase 1 construction complete (public Anderson-Shiro record). Package qualification: AEC-Q100 for Tesla, space screening for SpaceX. Only then cut over from Intel's existing U.S. packaging sites.                                                                                 |

Table 3. Packaging stand-up sequence aligned to announced dates.

Skipping Austin and going straight to Grimes contradicts the announced operating model.

## 9. What must not be assumed

The following are **not in the public record** and are not invented here:

- Packaging cleanroom area, tool counts, or wafer/package starts for the Grimes hall.
- Whether first Grimes packages are EMIB-only, EMIB-T, or Foveros Direct.
- HBM vendor, substrate vendor of record for Terafab, or ATE vendor.
- Whether Intel process runs in Terafab's building or Terafab remains a customer of Fab 9/Chandler during ramp.
- CHIPS Act money.
- Exact gas-plant MW or reservoir withdrawal (acre-feet). Those appear in TCEQ/USACE filings when filed; they are not in the press releases.

## 10. Failure modes already visible in this industry

1. **Hybrid-bond tool lead time** — Foveros Direct does not ramp if BESI/TEL slots are missed. Order against the 2028 date now.
2. **EMIB-T 90% → 98% yield** — public commentary: the last eight points are the hard ones. Plan scrap and rework for HBM (the expensive die).
3. **Substrate warpage on large AI packages** — Intel's own EMIB 3.5D brief exists because warpage and reticle limits are the constraint. Large-body substrates need in-line shadow-moiré, not end-of-line only.
4. **Utilities before tools** — a packaging bay with no UPW/CDA/chilled water is a warehouse. Gas plant and WWTP are on the county's critical path, not the bonder OEM's.
5. **Two reliability specs on one line** — automotive and space lots must be segregated (travelers, materials, outgassing). Mixing them is how you fail both qualifications.
6. **Export control** — orbital processors packaged in Texas still need a licensed technology-control plan before foreign-national process engineers (including Taiwan hires) touch the line.

## 11. Bottom line

Terafab's packaging facility is not a greenfield OSAT. It is an **Intel EMIB/Foveros (and EMIB-T) copy-exact line**, first stood up as a **closed-loop cell at Giga Texas**, then scaled at **Gibbons Creek** next to logic and memory, on **self-generated power and reservoir water**, qualified to **AEC-Q100 and space screening**.

Anything beyond that — bay size, tool list, first process of record — is not public and should be taken from Intel's POR and the TCEQ/USACE filings, not invented.

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